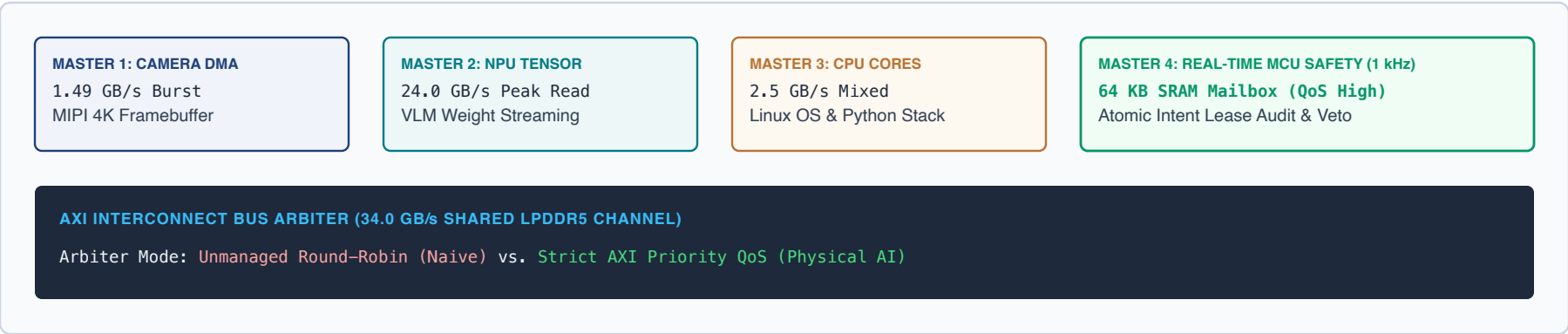


# UNIFIED MEMORY ARCHITECTURE (UMA) BUS CONTENTION & AXI QoS ARBITRATION

Guaranteed < 2 μs preemption for real-time safety mailboxes under heavy multi-master memory traffic



✗ UNMANAGED FIFO: MEMORY STARVATION & JITTER

DRAM Queue: [ Camera DMA ] [ Camera DMA ] [ NPU ] ✗ [ MCU Blocked ]  
Camera bursts lock out MCU safety access for up to 4.8 ms

- **Priority Inversion:** 1 kHz safety tick stalled by uncompressed video.
- **Inference Tail Spikes:** NPU weight fetch stalls \$implies\$ P99 explodes to 135 ms.
- **Physical Breach:** Exceeds dynamic stability deadline \$implies\$ Hardware E-Stop.

✓ STRICT AXI QoS: DEDICATED BANDWIDTH & < 2 μs PREEMPTION

DRAM Queue: [ MCU Safety ★ (QoS=15) ] → Instant Priority Bypass  
Preempts background video DMA in < 2 microseconds

- **Hardware QoS Priority Tags:** MCU Mailbox (QoS=15), NPU (QoS=10), Camera (QoS=4).
- **Zero Copy Ring Buffer:** Eliminates 3x user/kernel memory copying.
- **Deterministic Metrology:** Tail latency bounded at **P99 = 24.2 ms << τ<sub>world</sub>**.